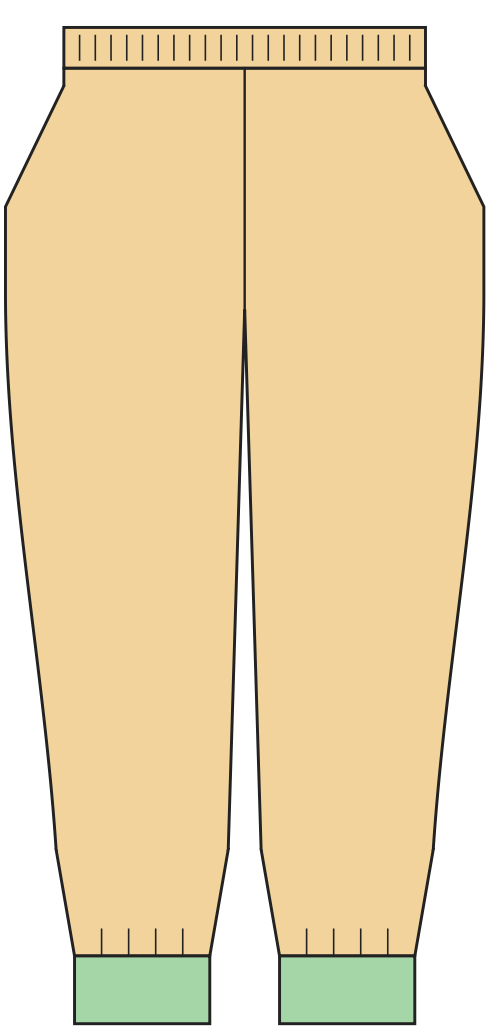
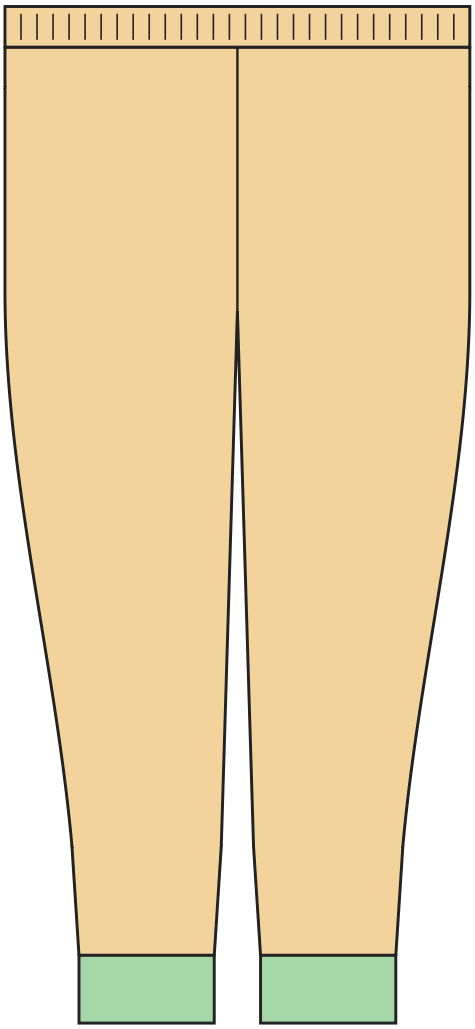
Joggers

size 122 · normal fit, Sweat / French Terry (25 % stretch)

6 parts · 25 steps · medium · about 3.5 h



Front view



Back view

Sewn with: Sewing machine



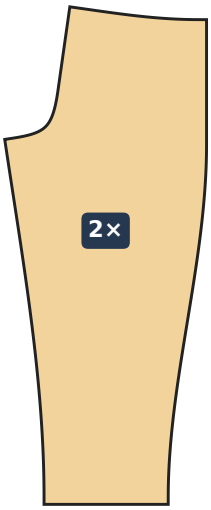
Sewing machine



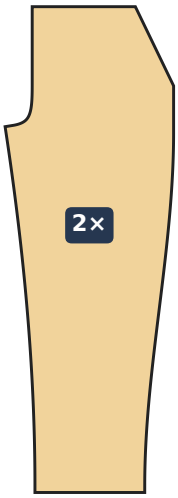
Iron

Read the steps once before you start. Every picture is drawn from your pattern pieces. Sew a test garment from cheap fabric first and adjust before you cut the good fabric.

Parts



B · Back leg



A · Front leg



C · Side panel (outer pocket bag)



D · Inner pocket bag



F · Leg cuff



E · Waistband (elastic) (cut on fold)

Seam allowance (included): 1.0 cm

Material

Main fabric
0.90 m at 150 cm width

Rib knit
0.20 m at 70 cm width

Elastic
Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Sewing machine
narrow zigzag 1 × 2.5 mm (or overlocker); twin needle 4 mm, stitch 3 mm; longest straight stitch, no backstitch; wide zigzag 4 × 2.5 mm

Iron
steam

Polyester sewing thread
In the colour of the fabric.

Pins or clips
Pin across the seam line.

Chalk or a fabric pen
For notches, labels and placement lines.

Colours in this booklet

Main fabric
right side · wrong side

Rib knit
right side · wrong side

Sew now (machine)

Sew by hand

Already sewn

Fold / press line

Marking (chalk)

Pin

Notch: match them

Right/wrong side of the fabric = the side of the fabric. Left/right (as worn) = seen from the child wearing it: the left front leg is on the right when you look at the child.

Basics: Sewing machine

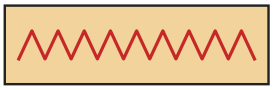
Steps are written for a sewing machine with zigzag. Have an overlocker or a coverstitch machine? Tell the generator which machines you own and the steps use them.

Basics: the sewing machine



Straight stitch

Seams in woven fabric, 2.5 mm; backstitch at the start and the end.



Zigzag

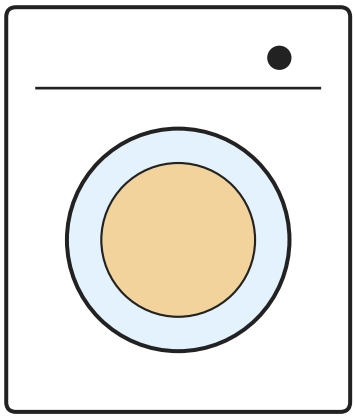
Knit seams (narrow) and finishing the edges (wide).



Topstitching

Right side up, with the same distance to the edge all the way; use the edge of the presser foot as a guide.

1 Pre-wash the fabric



30–40°

1. Wash and dry it the way you will wash the garment later, so it shrinks now, not after sewing.

2 Cut out all pieces

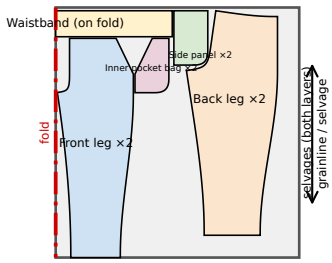


Scissors

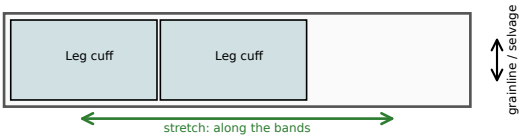
1. Follow the cutting layout
2. Grainline arrows parallel to the selvage.
3. Rib knit: the ribs run along the grainline, so every band stretches along its length (across the fabric).

Tip: Cut both layers at once with long strokes; keep the fabric flat on the table.

Main fabric · 150 cm · folded (double layer)



Rib knit · 70 cm · single layer

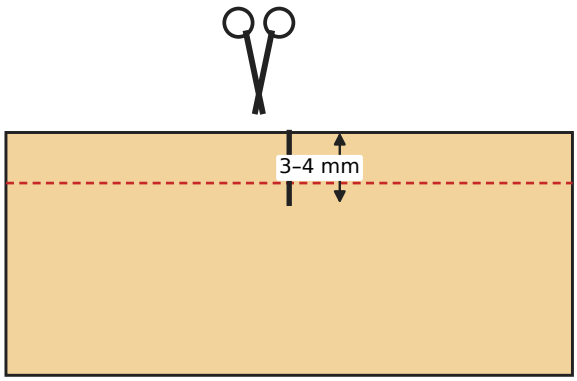


3 Snip the notches

Chalk

1. Cut 3–4 mm into the seam allowance at every notch.

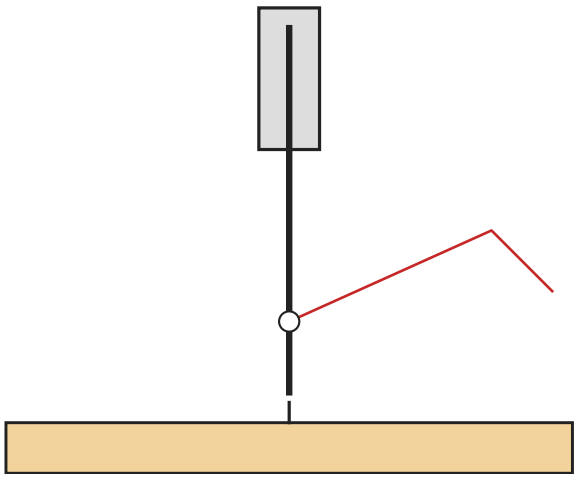
⚠ Important: Never deeper than half the seam allowance.



4 Set up the machine

1. Sewing machine: jersey/stretch needle 75–80, polyester thread.
2. Sew a test seam on two layers of a scrap.

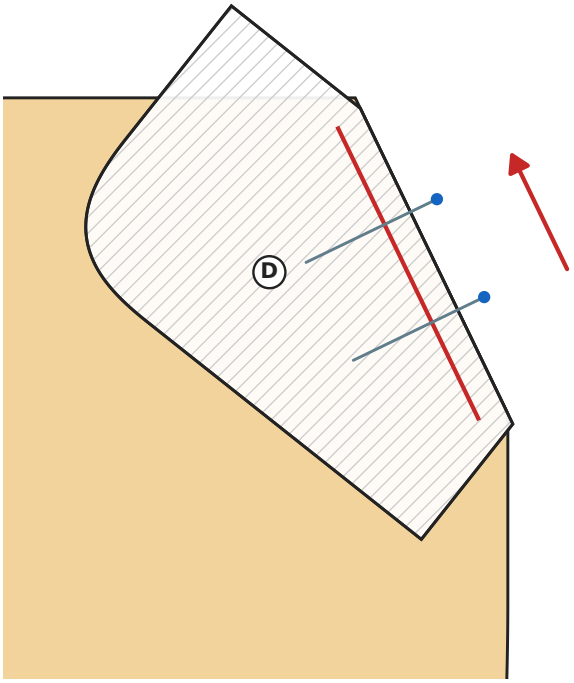
Tip: See the basics page at the front of the booklet.



5 Sew the pocket bag to the opening

Sewing machine

Pins



1. Lay the inner pocket bag right sides together on the pocket opening of the front leg, edges level, and pin.
2. Sew along the opening with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: The opening is on the bias: do not stretch it.

Tip: A strip of stay tape ironed onto the opening keeps it in shape.

2×

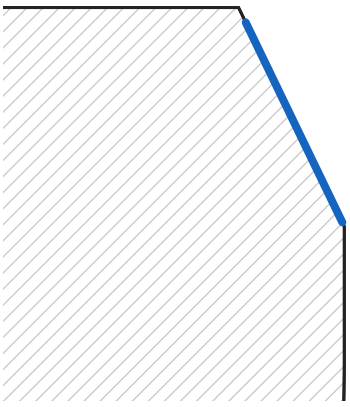
6 Clip and turn the bag inside

Iron

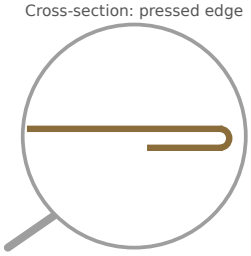
1. Clip the curved allowance every 1 cm, close to the stitching.
2. Turn the bag to the inside and press the edge.

Iron: steam

⚠ Important: Roll the edge 1–2 mm to the inside: no pocket bag may show from the outside.



Cross-section: pressed edge



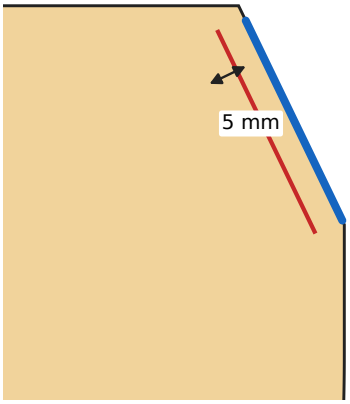
2×

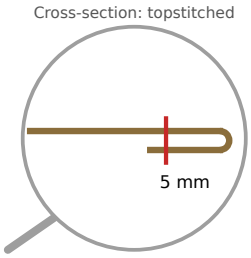
7 Topstitch the pocket edge

Sewing machine

1. Right side up, topstitch 5 mm from the edge through all layers.

Sewing machine: twin needle 4 mm, stitch 3 mm

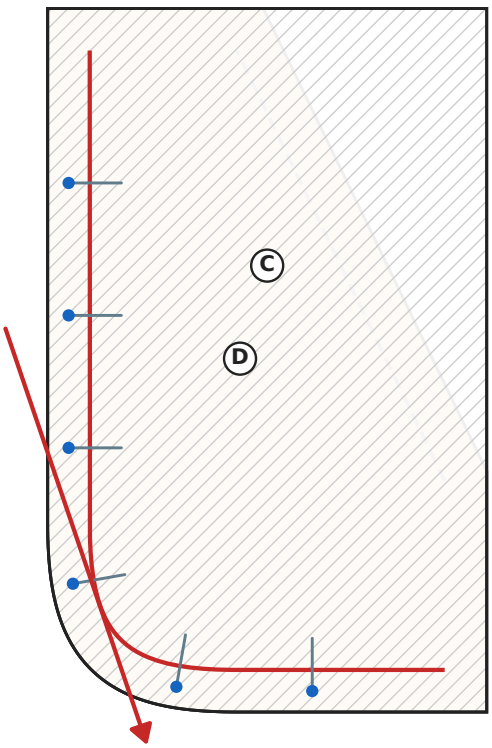




2×

8 Sew the pocket bag seam

Sewing machine · Pins



1. Lay the side panel right sides together under the pocket bag so that the "pocket opening" mark meets the opening edge.
2. Pin the two pocket bag layers and sew them together along the curve with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

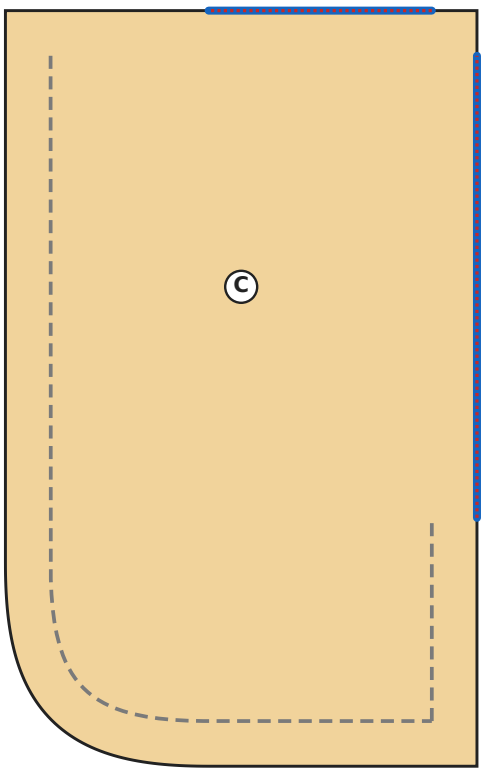
⚠ Important: Fold the front leg out of the way: sew only the two pocket bag layers.

Tip: Turn it over afterwards and check from the outside that the opening lies flat.

2×

9 Baste the side panel to the leg

Sewing machine



1. Lay the leg flat on the side panel.
2. Baste the top edge and the side edge together, inside the seam allowance.

Sewing machine: longest straight stitch, no backstitch

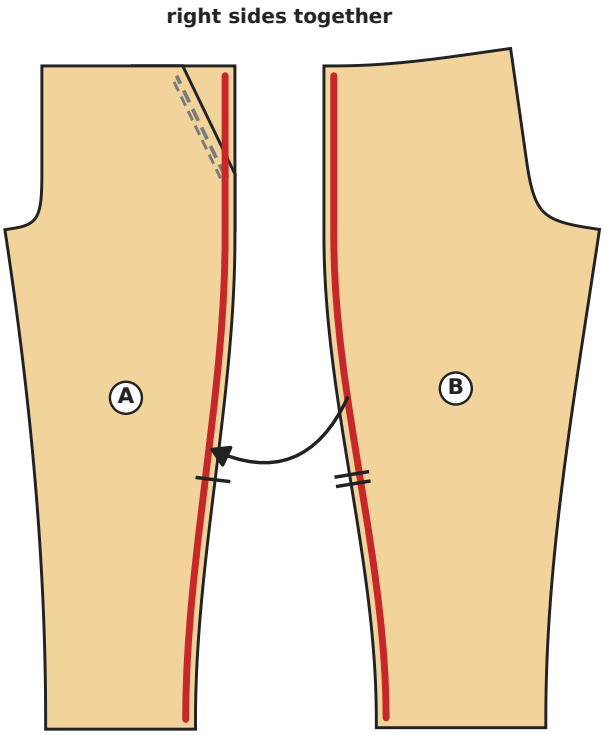
⚠ Important: The pocket opening lies flat, not pulled.

Tip: Baste 4 mm from the edge, inside the seam allowance.

2×

10 Sew the side seams

Sewing machine · Pins



1. Lay front and back of the same leg right sides together, side edges level: single notches = front, double notches = back.
2. Pin, then sew with 10 mm seam allowance from the top edge to the hem.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

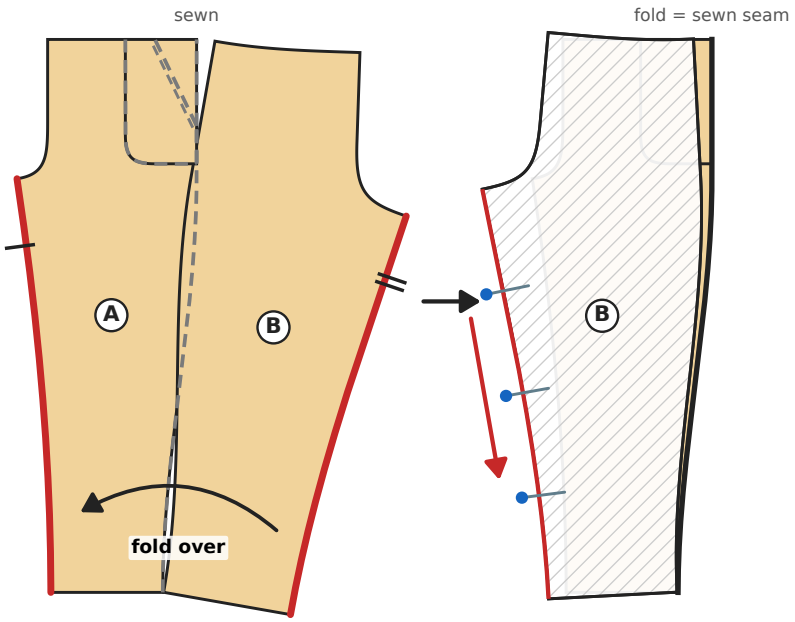
⚠ Important: Always the front and back of the SAME leg (FL + BL, FR + BR).

2×

11 Sew the inseams

Sewing machine

Pins



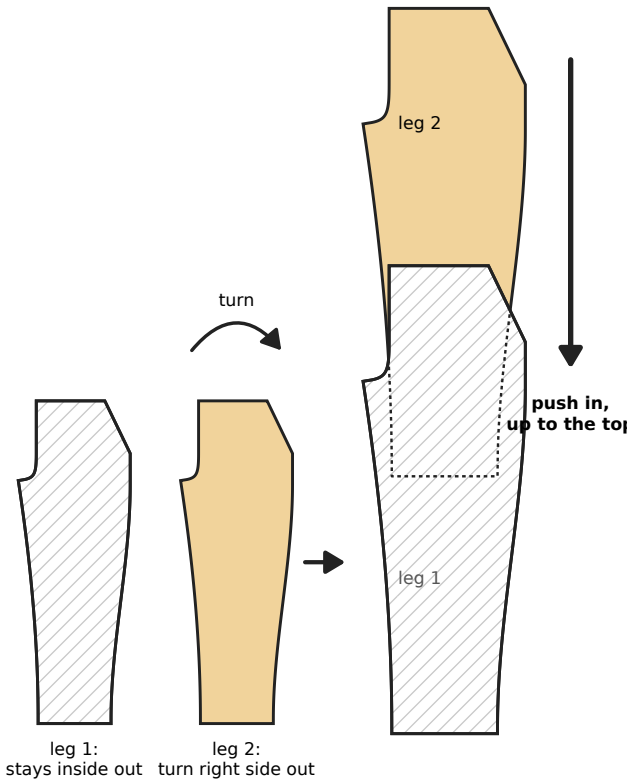
1. Open the leg out at the side seam and fold it right sides together, inseam edges level.
2. Pin the notches first, then the ends.
3. Sew with 10 mm seam allowance from the crotch point to the hem.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Start exactly at the crotch point: the four seams meet there later.

2×

12 Put one leg inside the other

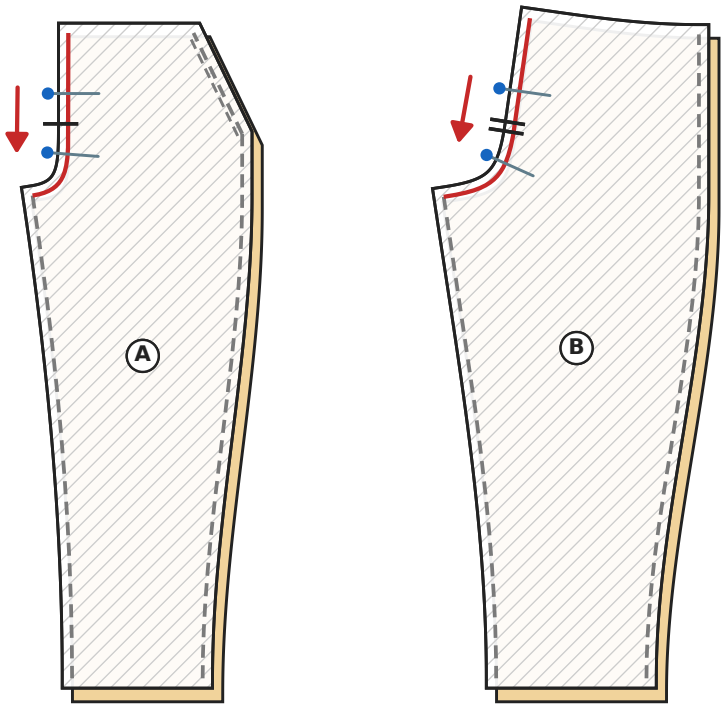


1. Turn leg 2 right side out.
2. Leg 1 stays wrong side out.
3. Push leg 2 into leg 1 until the top edges are even
4. The hems point the same way.

Tip: Now the legs lie right sides together and the crotch curves lie exactly on top of each other.

13 Sew the crotch seam

Sewing machine · Pins



1. Pin the crotch curves together, starting at the inseams: they lie exactly on top of each other.
2. Sew with 10 mm seam allowance in one go from the front top edge to the back top edge.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

⚠ Important: Sew the crotch curve a second time for strength.

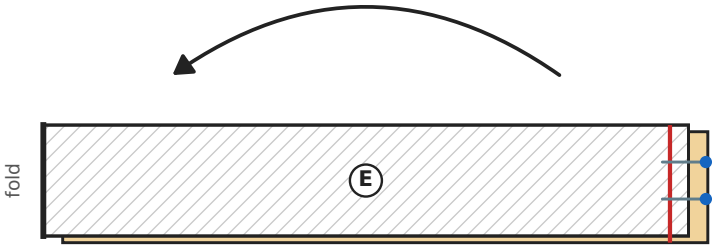
Tip: Press the inseam allowances in opposite directions: the crossing lies flat. Pull the inner leg out afterwards.

14 Close the waistband into a ring

Sewing machine · Pins

1. Fold the waistband crosswise right sides together and sew the short ends (centre back) with 10 mm seam allowance
2. Leave 3 cm open for the elastic.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

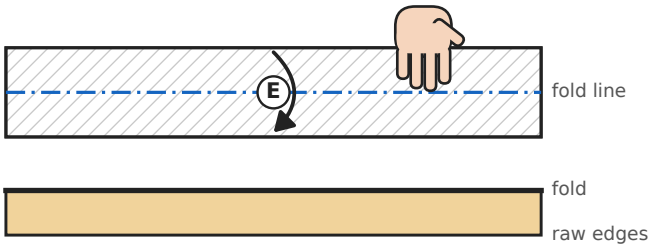


15 Fold the waistband in half

Iron

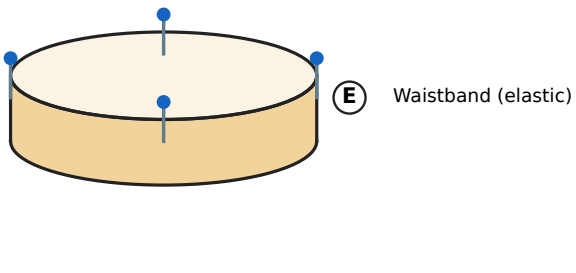
1. Fold lengthwise wrong sides together along the fold line and press.

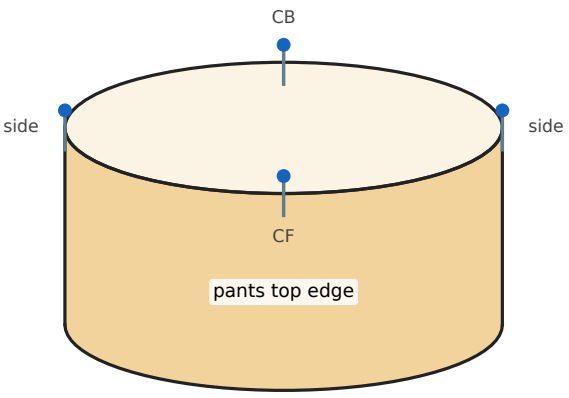
Iron: steam



16 Mark quarters

1. Mark centre front, centre back and both sides with pins, on the waistband and on the pants top edge.





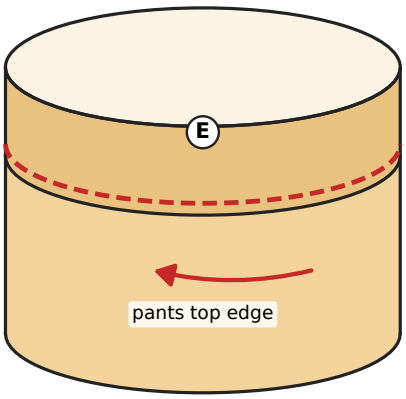
17 Sew the waistband on

Sewing machine

Pins

1. Pants right side out: slide the waistband over them, raw edges level, the opening at the back on the inside.
2. Match the quarter pins.
3. Sew all round with 10 mm seam allowance.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



waistband folded, raw edges down

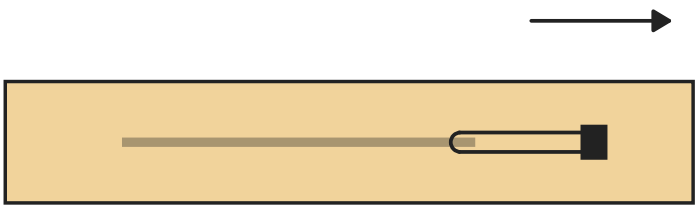
18 Thread the elastic

Sewing machine

Safety pin

1. With a safety pin through the opening
2. Overlap the ends by 2 cm and sew them together.

Sewing machine: wide zigzag 4 × 2.5 mm



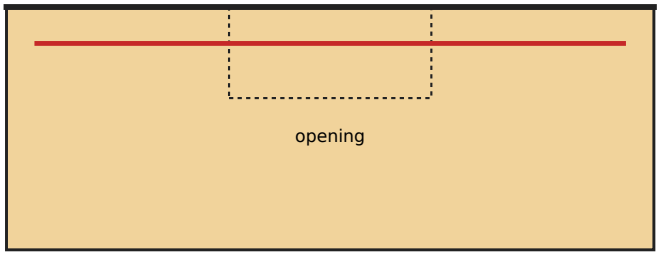
elastic, with a safety pin

19 Close the opening

Sewing machine

1. Spread the gathers evenly, close the opening and stitch in the ditch at the side seams so the elastic cannot twist.

Sewing machine: twin needle 4 mm, stitch 3 mm



stitch close to the fold, over the opening

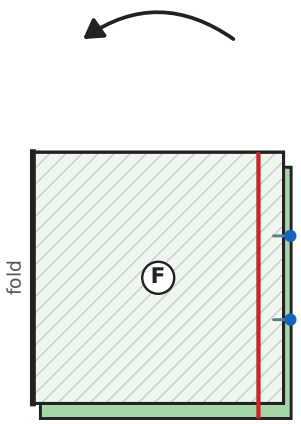
20 Close the cuffs into rings

Sewing machine

Pins

1. Fold each cuff crosswise right sides together, short edges level.
2. Sew with 10 mm seam allowance
3. Finger-press the allowances open.

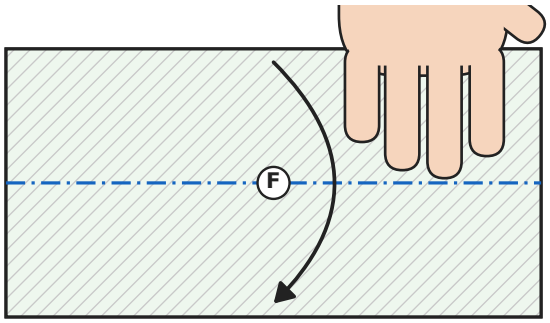
Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)



2×

21 Fold the cuffs in half

Iron



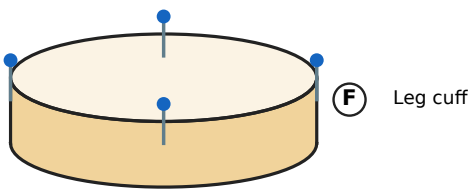


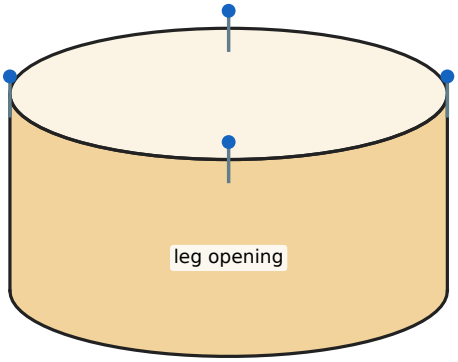
1. Fold the ring lengthwise wrong sides together, raw edges level.
2. The seam allowance is now inside
3. The right side shows all round.

Iron: steam

2×

22 Mark quarters





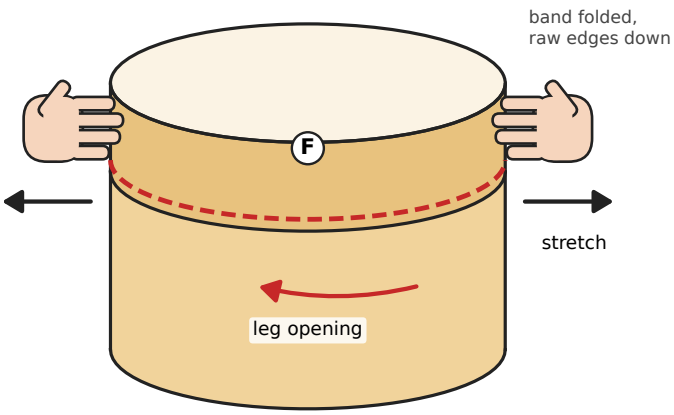
1. Mark quarters with pins on the cuff and on the leg opening
2. The cuff seam is one of them.

2×

23 Sew the cuffs on

Sewing machine

Pins



1. Turn the leg right side out.
2. Slide the cuff over the leg end, raw edges level, cuff seam on the inseam.
3. Match the quarter pins.
4. Sew with 10 mm seam allowance, cuff on top, stretching only the cuff.

Sewing machine: narrow zigzag 1 × 2.5 mm (or overlocker)

Tip: One hand in front of the presser foot, one behind it: both stretch evenly.

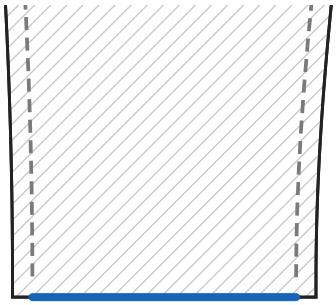
2×

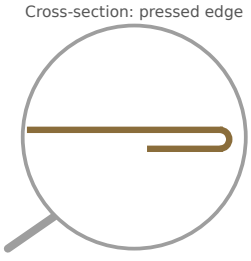
24 Press the allowance up

Iron

1. Press the seam allowance up towards the pants.

Iron: steam

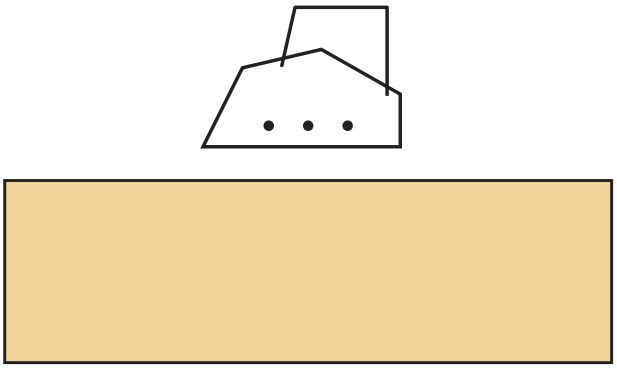




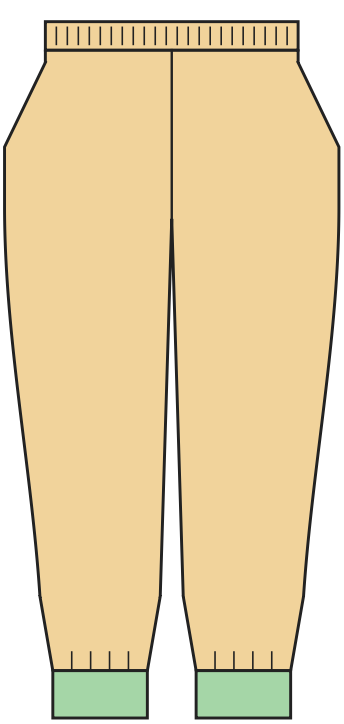
2×

25 Final pressing

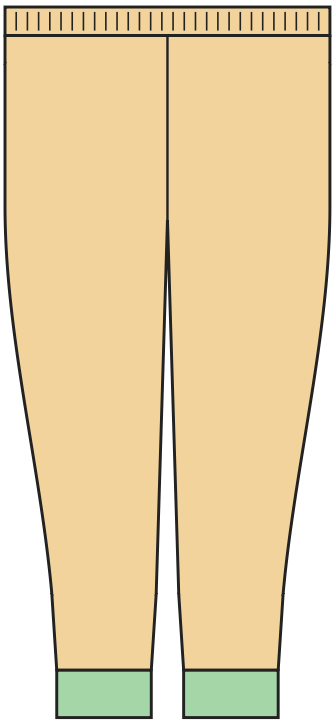
1. Press all seams with steam
2. Do not press down hard on sweat.



Done!



Front view



Back view

After the test garment: Try the test garment on the child: length, width at the hip, the crotch and the waistband. Adjust the measurements or options and print the pattern again before you cut the good fabric.

Joggers · sizes 98 · 110 · 122

normal fit, Sweat / French Terry (25 % stretch) · original parametric draft. The pattern is yours to use.

> **Important:** With a new pattern, first sew a **test garment** from inexpensive fabric with similar stretch and check the fit.

Files

- PDF roll 610 mm (600 × 1017 mm): `joggers_98_110_122_roll610.pdf`
- Preview roll 610 mm: `joggers_98_110_122_roll610_preview.png`
- SVG master 1:1: `joggers_98_110_122_master_1to1.svg`
- Cutting layout: `cutting_layout.png`
- Sewing instructions (PDF): `sewing_instructions.pdf`
- Sewing instructions (Markdown): `sewing_instructions.md`

Printing

- Always print at **100 % / "Actual size"** (not "Fit to page").
- Measure the **5 × 5 cm** control square before you cut.
- Plotter (610 mm roll): the page is 600 mm wide in portrait. In the print dialog choose a custom paper size with exactly this width and length, and turn rotation/scaling off.

Body measurements

Body measurement	Value	Source
Body height	122.0 cm	table
Chest	62.0 cm	table
Waist	56.0 cm	table
Hip	66.0 cm	table
Head	52.0 cm	table
Neck	28.5 cm	table
Inseam	54.0 cm	table
Side length	72.0 cm	table
Arm length	47.0 cm	table
Back length	29.5 cm	table
Shoulder width	10.0 cm	table
Thigh	37.0 cm	table
Ankle	17.8 cm	table

Finished measurements (virtually sewn)

Finished measurement	Value
Hip width	72.6 cm
Thigh width	46.4 cm
Leg width at mid-leg	36.0 cm
Leg opening	20.0 cm
Waistband relaxed	50.4 cm
Waistband stretched (pants top edge)	69.8 cm
Outseam	73.5 cm
Inseam	54.0 cm
Front rise	19.5 cm
Back rise	21.1 cm
Waistband height	3.0 cm
Cuff height	5.0 cm
Crotch seam total	46.9 cm

Sizes in this pattern (nested)

All sizes are drawn on top of each other, aligned at the same construction point. Every size has its own colour and line and its own **PDF layer**: in Acrobat/Reader open the Layers panel and switch off the sizes you do not need before printing. Cut along the line of your size. Sizes can be blended: e.g. width of one size, length of another, by drawing a smooth line between the two outlines.

Size	Line	Fabric needed	Hip width	Thigh width	Leg width at mid-leg	Leg opening
size 98	red, solid	Main fabric: 0.65 m; Rib knit: 0.15 m	62.7 cm	39.9 cm	31.3 cm	17.2 cm
size 110	blue, long dashes	Main fabric: 0.80 m; Rib knit: 0.15 m	67.1 cm	42.8 cm	33.3 cm	18.4 cm
size 122	green, short dashes	Main fabric: 0.90 m; Rib knit: 0.20 m	72.6 cm	46.4 cm	36.0 cm	20.0 cm

Fabric, finished measurements and sewing steps below refer to the largest size (size 122); the validation report has one table per size.

Pattern pieces

Piece	Cutting	Fabric	Seam allowance
Front leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Back leg	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Side panel (outer pocket bag)	cut 2 (mirrored pair)	Main fabric	seam allowance 1.0 cm included
Inner pocket bag	cut 2 (mirrored pair)	Main fabric or cotton jersey	seam allowance 1.0 cm included
Waistband (elastic)	cut 1 on fold	Main fabric	seam allowance 1.0 cm included
Leg cuff	cut 2	Rib knit	seam allowance 1.0 cm included

Fabric needed

- **Main fabric:** about 0.90 m at 150 cm width (folded selvage to selvage, cut on fold pieces at the fold, incl. 10 % for shrinkage; single layer would need 0.90 m)
- **Rib knit:** about 0.20 m at 70 cm width (single layer, incl. 10 % for shrinkage)
- Elastic 20 mm wide, length 50 cm (+2 cm overlap for joining)

Cutting layout: see `cutting_layout.png` (grainline arrow parallel to the selvage, "flipped" = place the piece face down, "×2" = cut through both layers of the folded fabric, which gives a mirrored pair, "(unfolded)" = cut-on-fold piece shown opened out).

Cut on fold (Waistband (elastic)): the paper piece is only one half. Its fold edge (dash-dot line, bracket arrows "PLACE ON FOLD") is the centre of the piece: it gets no seam allowance and is never cut. On the folded fabric, lay the fold edge exactly on the fabric fold, pin it and cut through both layers; opened out you get the whole piece. Where the cutting layout shows a piece "(unfolded)" on a single layer, fold the fabric along the grain just wide enough for the piece and place the fold edge on that fold, or trace the paper piece, flip it over along the fold edge and trace the second half.

Validation report

Sewability check · size 98

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	51.4 cm	51.4 cm	+0.0 mm	✓ OK	
Seam pairs	Inseam front ↔ back	35.9 cm	35.9 cm	+0.0 mm	✓ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	17.2 cm	17.2 cm	+0.0 mm	✓ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	20.8 cm	20.8 cm	+0.0 mm	✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Pocket opening: front leg ↔ pocket bag	10.5 cm	10.5 cm	+0.0 mm	✓ OK	
Seam pairs	Pocket bag seam: pocket bag ↔ side panel	19.4 cm	19.4 cm	+0.0 mm	✓ OK	
Seam pairs	Pocket bag ↔ side panel at the side seam	3.6 cm	3.6 cm	+0.0 mm	✓ OK	
Seam pairs	Waistband ↔ pants top edge	59.9 cm	59.9 cm	+0.0 mm	✓ OK	
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 24.6 cm	17.2 cm = 70 %	-73.9 mm	✓ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	✓ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.8°	+0.8°	✓ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband at the side panel	90°	90.0°	+0.0°	✓ OK	
Common practice	Hip ease for fit 'normal'	5 – 18 %	10 %	+0 %	✓ OK	guide value for this fit
Measurements	Hip circumference pattern = body + ease	62.7 cm	62.7 cm	+0.0 %	✓ OK	body 57.0 cm + +10 % ease
Measurements	Top edge fits over the hip	≥ 58.1 cm	68.9 cm	+107.6 mm	✓ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	11.8 cm – 20.9 cm	18.5 cm	+0.0 mm	✓ OK	body rise 17.0 cm
Measurements	Inseam length plausible	28.9 cm – 44.8 cm	39.7 cm	+0.0 mm	✓ OK	body inseam 40.0 cm
Measurements	Foot fits through the leg opening	≥ 20.8 cm	25.5 cm	+47.3 mm	✓ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Side panel (outer pocket bag): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included

Area	Check	Target	Actual	Δ	Result	Note
Labels	Side panel (outer pocket bag): grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Inner pocket bag: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Inner pocket bag: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric or cotton jersey
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		✓ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		✓ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		✓ OK	
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		✓ OK	

Result: 39 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 627}}`).

Name	Value	Derived from	Note
`hip_width`	62.7 cm	hip × (1 + 10 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	15.2 cm	hip_width / 4 − front_back_offset	
`back_width`	16.2 cm	hip_width / 4 + front_back_offset	
`front_crotch_extension`	3.1 cm	5 % of hip_width	
`back_crotch_extension`	6.3 cm	10 % of hip_width	
`rise`	15.5 cm	(side length − inseam) + seat ease − waistband height	
`inseam_finished`	39.7 cm	inseam − allowance to the floor + blousing	
`hem_width`	24.6 cm	cuff length / cuff stretch ratio	

Sewability check · size 110

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	59.0 cm	59.0 cm	+0.0 mm	✓ OK	
Seam pairs	Inseam front ↔ back	43.5 cm	43.5 cm	+0.0 mm	✓ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	17.3 cm	17.3 cm	+0.0 mm	✓ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	21.1 cm	21.1 cm	+0.0 mm	✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Pocket opening: front leg ↔ pocket bag	10.6 cm	10.6 cm	+0.0 mm	✓ OK	
Seam pairs	Pocket bag seam: pocket bag ↔ side panel	20.4 cm	20.4 cm	+0.0 mm	✓ OK	
Seam pairs	Pocket bag ↔ side panel at the side seam	4.1 cm	4.1 cm	+0.0 mm	✓ OK	
Seam pairs	Waistband ↔ pants top edge	64.3 cm	64.3 cm	+0.0 mm	✓ OK	
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 26.4 cm	18.4 cm = 70 %	-79.1 mm	✓ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	✓ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.7°	+0.7°	✓ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband at the side panel	90°	90.0°	+0.0°	✓ OK	
Common practice	Hip ease for fit 'normal'	5 – 18 %	10 %	+0 %	✓ OK	guide value for this fit
Measurements	Hip circumference pattern = body + ease	67.1 cm	67.1 cm	+0.0 %	✓ OK	body 61.0 cm + +10 % ease
Measurements	Top edge fits over the hip	≥ 62.2 cm	73.9 cm	+117.3 mm	✓ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	13.2 cm – 22.3 cm	18.5 cm	+0.0 mm	✓ OK	body rise 17.0 cm
Measurements	Inseam length plausible	34.7 cm – 53.8 cm	47.9 cm	+0.0 mm	✓ OK	body inseam 48.0 cm
Measurements	Foot fits through the leg opening	≥ 22.2 cm	27.3 cm	+50.7 mm	✓ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Side panel (outer pocket bag): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included

Area	Check	Target	Actual	Δ	Result	Note
Labels	Side panel (outer pocket bag): grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Inner pocket bag: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Inner pocket bag: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric or cotton jersey
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		✓ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		✓ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		✓ OK	
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		✓ OK	

Result: 39 OK, 0 warning(s), 0 error(s): the pattern is sewable.

Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 671}}`).

Name	Value	Derived from	Note
`hip_width`	67.1 cm	hip × (1 + 10 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	16.3 cm	hip_width / 4 − front_back_offset	
`back_width`	17.3 cm	hip_width / 4 + front_back_offset	
`front_crotch_extension`	3.4 cm	5 % of hip_width	
`back_crotch_extension`	6.7 cm	10 % of hip_width	
`rise`	15.5 cm	(side length − inseam) + seat ease − waistband height	
`inseam_finished`	47.9 cm	inseam − allowance to the floor + blousing	
`hem_width`	26.4 cm	cuff length / cuff stretch ratio	

Sewability check · size 122

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Side seam front ↔ back	65.7 cm	65.7 cm	+0.0 mm	✓ OK	
Seam pairs	Inseam front ↔ back	49.2 cm	49.2 cm	+0.0 mm	✓ OK	
Seam pairs	Front crotch seam (left ↔ right piece)	18.4 cm	18.4 cm	+0.0 mm	✓ OK	
Seam pairs	Back crotch seam (left ↔ right piece)	22.5 cm	22.5 cm	+0.0 mm	✓ OK	

Area	Check	Target	Actual	Δ	Result	Note
Seam pairs	Pocket opening: front leg ↔ pocket bag	11.4 cm	11.4 cm	+0.0 mm	✓ OK	
Seam pairs	Pocket bag seam: pocket bag ↔ side panel	22.0 cm	22.0 cm	+0.0 mm	✓ OK	
Seam pairs	Pocket bag ↔ side panel at the side seam	4.5 cm	4.5 cm	+0.0 mm	✓ OK	
Seam pairs	Waistband ↔ pants top edge	69.8 cm	69.8 cm	+0.0 mm	✓ OK	
Seam pairs	Leg cuff ↔ leg opening (stretched)	70–85 % of 28.5 cm	20.0 cm = 70 %	-85.5 mm	✓ OK	rib band is stretched while sewing
Angles/transitions	Centre front ↔ waistband	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Centre back ↔ waistband	90°	90.0°	-0.0°	✓ OK	
Angles/transitions	Crotch ↔ inseam (front + back)	180°	180.7°	+0.7°	✓ OK	
Angles/transitions	Inseam ↔ hem front	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Hem ↔ side seam front	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Inseam ↔ hem back	90°	90.0°	+0.0°	✓ OK	
Angles/transitions	Hem ↔ side seam back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband back	90°	90.1°	+0.1°	✓ OK	
Angles/transitions	Side seam ↔ waistband at the side panel	90°	90.0°	+0.0°	✓ OK	
Common practice	Hip ease for fit 'normal'	5 – 18 %	10 %	+0 %	✓ OK	guide value for this fit
Measurements	Hip circumference pattern = body + ease	72.6 cm	72.6 cm	+0.0 %	✓ OK	body 66.0 cm + +10 % ease
Measurements	Top edge fits over the hip	≥ 67.3 cm	80.3 cm	+129.4 mm	✓ OK	top edge incl. usable stretch ≥ hip
Measurements	Front rise plausible	14.6 cm – 24.0 cm	19.5 cm	+0.0 mm	✓ OK	body rise 18.0 cm
Measurements	Inseam length plausible	39.0 cm – 60.5 cm	54.0 cm	+0.0 mm	✓ OK	body inseam 54.0 cm
Measurements	Foot fits through the leg opening	≥ 24.1 cm	29.5 cm	+54.8 mm	✓ OK	heel-instep circumference ≈ 1.35 × ankle
Geometry	Front leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Front leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Back leg: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Back leg: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Side panel (outer pocket bag): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included

Area	Check	Target	Actual	Δ	Result	Note
Labels	Side panel (outer pocket bag): grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric
Geometry	Inner pocket bag: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Inner pocket bag: grainline & cutting	present	cut 2 (mirrored pair)		✓ OK	Main fabric or cotton jersey
Geometry	Waistband (elastic): outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Waistband (elastic): grainline & cutting	present	cut 1 on fold		✓ OK	Main fabric
Geometry	Leg cuff: outline & seam allowance	closed, SA present	ok		✓ OK	seam allowance 1.0 cm included
Labels	Leg cuff: grainline & cutting	present	cut 2		✓ OK	Rib knit
Geometry	Waistband (elastic): cut-on-fold edge	straight, no SA, 90° corners, on grain	ok, corners ≤ 0.0° off 90°		✓ OK	Centre front
Labels	Front leg: notches	on the cutting line	3 notch mark(s)		✓ OK	
Labels	Back leg: notches	on the cutting line	6 notch mark(s)		✓ OK	

Result: 39 OK, 0 warning(s), 0 error(s): the pattern is sewable.

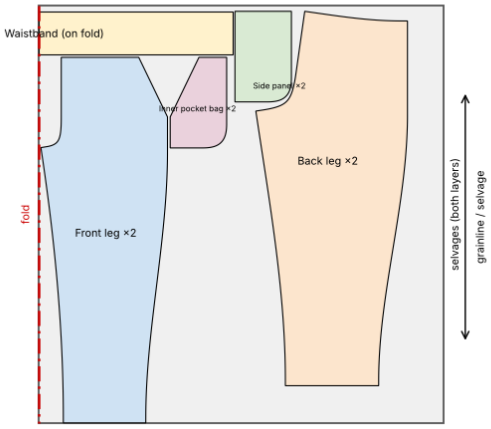
Construction values

Override any of these with the option `construction`, values in mm (e.g. `{"construction": {"hip_width": 726}}`).

Name	Value	Derived from	Note
`hip_width`	72.6 cm	hip × (1 + 10 % ease)	
`front_back_offset`	0.5 cm	front quarter narrower, back quarter wider by this amount	
`front_width`	17.7 cm	hip_width / 4 − front_back_offset	
`back_width`	18.7 cm	hip_width / 4 + front_back_offset	
`front_crotch_extension`	3.6 cm	5 % of hip_width	
`back_crotch_extension`	7.3 cm	10 % of hip_width	
`rise`	16.5 cm	(side length − inseam) + seat ease − waistband height	
`inseam_finished`	54.0 cm	inseam − allowance to the floor + blousing	
`hem_width`	28.5 cm	cuff length / cuff stretch ratio	

Cutting layout · Joggers size 122

Main fabric: 150 cm wide, folded (double layer)
needed ≈ 0.90 m



Rib knit: 70 cm wide, single layer
needed ≈ 0.20 m

